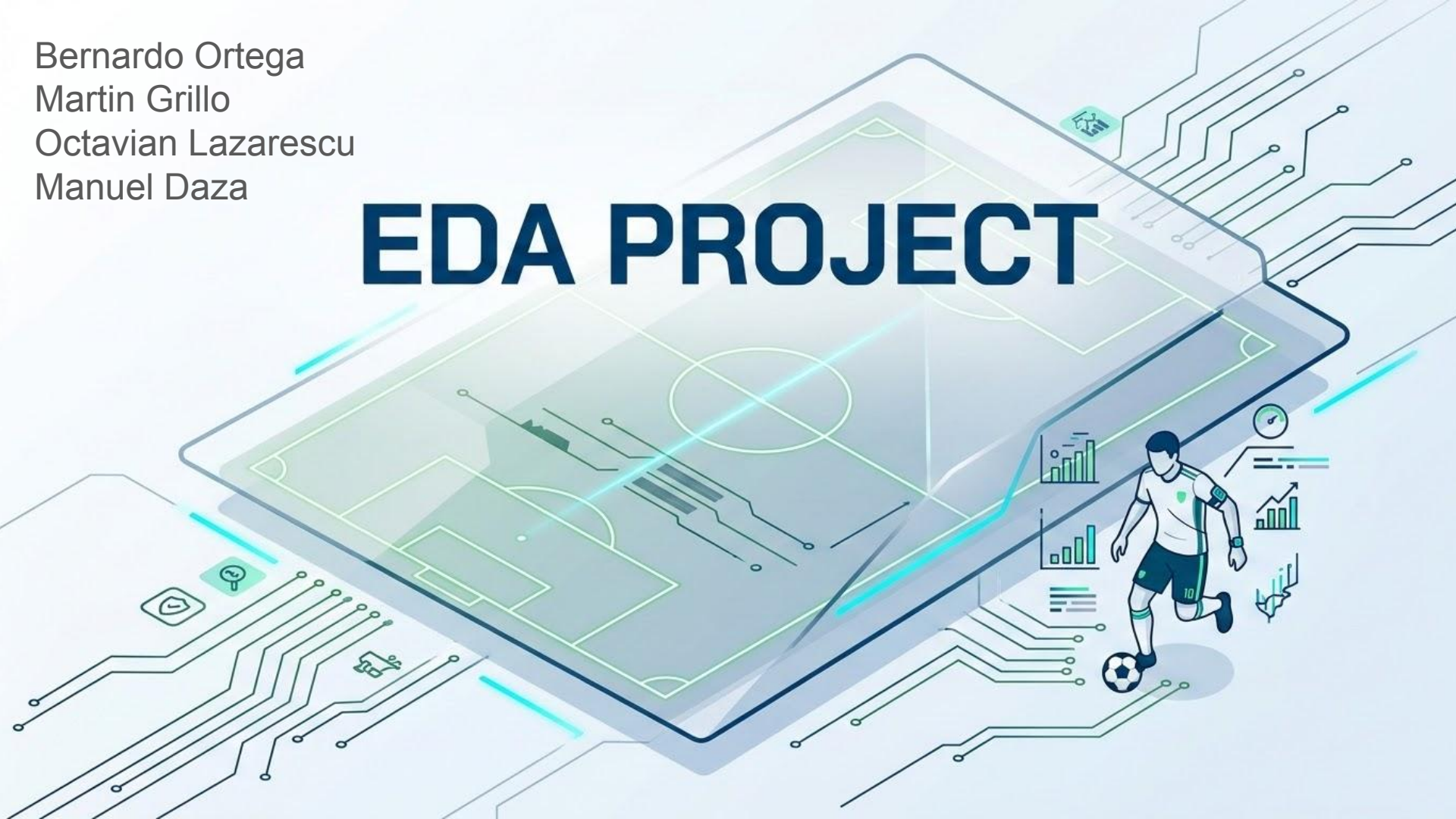


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EDA PROJECT



INDEX

- Group Organization
- Github
- How accurately do Expected Goals (xG) predict the final match result?
- Does recent team performance influence match outcomes?
- Do teams with more current points consistently achieve better match results?
- How do team player ratings and team formations influence match performance?

Organization

github: distinct repository <https://github.com/magrimuc/prj-2-eda-sql-football-matches-statistics>

containing:

- README.md
- at least 1 notebook each team member, one master notebook
- .pdf s with detailed tasks (roadmap, project plan) → still uneasy to assign tasks :)
- uneasy: the dataset (too large → .gitignore) - cleaned dataset was small enough

Q1 How accurately do Expected Goals (xG) predict the final match result?

Main Variables from the dataset

total_xg

Sum of expected goals from both teams in a match. Represents the total quality of chances created.

total_goals

Sum of actual goals scored by both teams. This is the outcome we're trying to predict.

3. xg_difference

Difference in xG between home and away teams ($\text{home_xg} - \text{away_xg}$). Measures which team was expected to dominate the match.

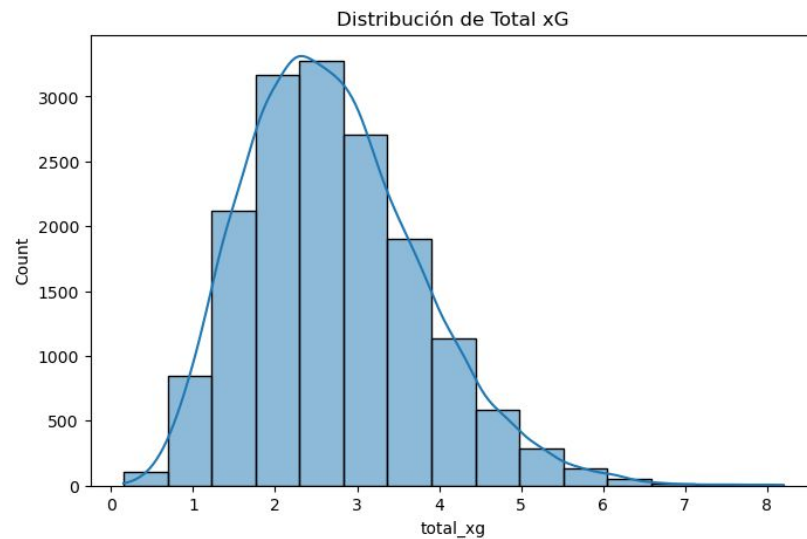
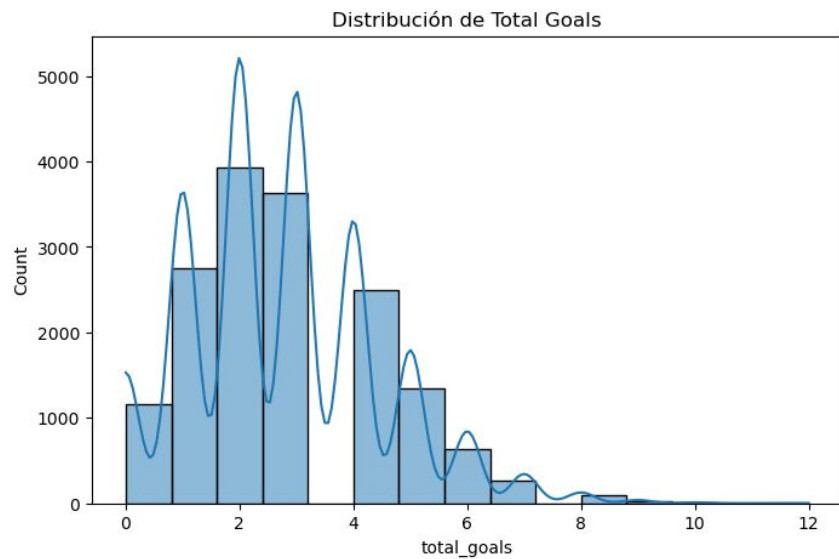
4. goal_difference

Actual goal difference ($\text{home_score} - \text{away_score}$). Measures which team actually dominated the scoreline.

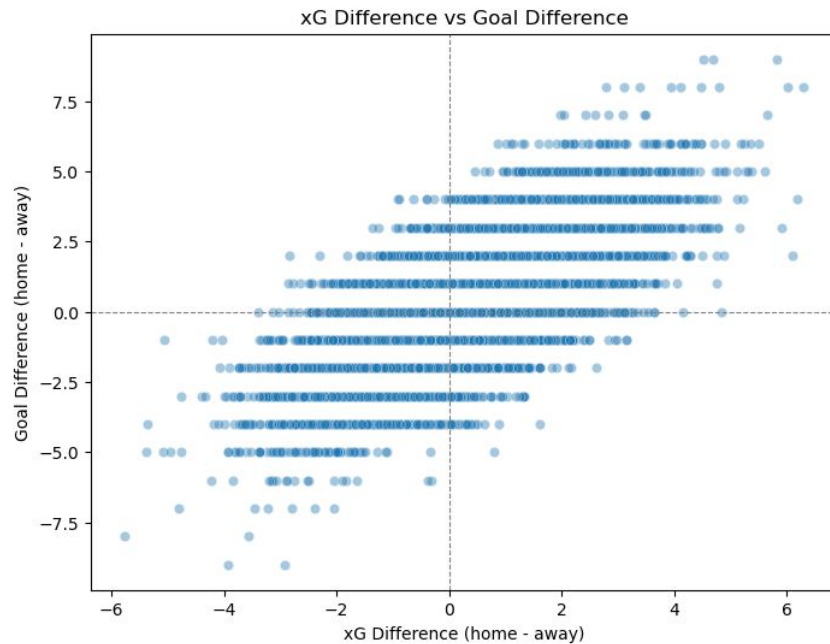
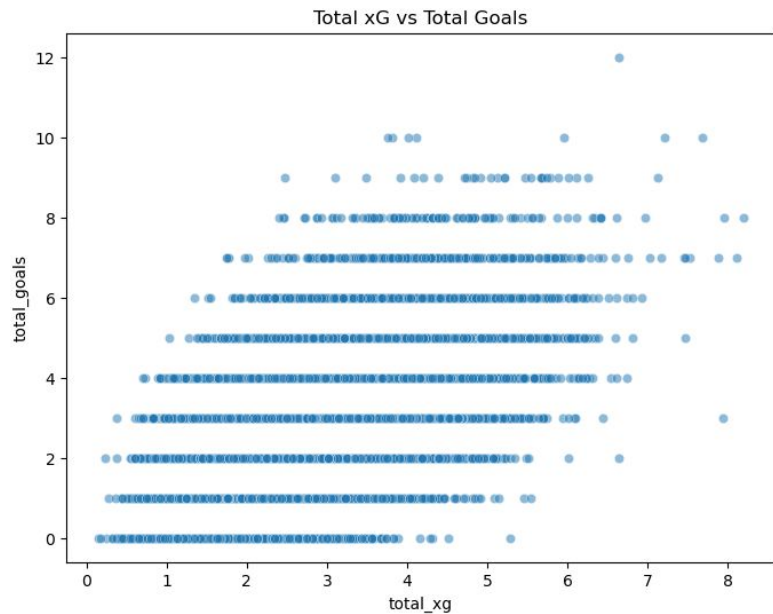
5. match_result

Categorical outcome (Home Win / Away Win / Draw), used to check whether xG relates to the final result, not just the score.

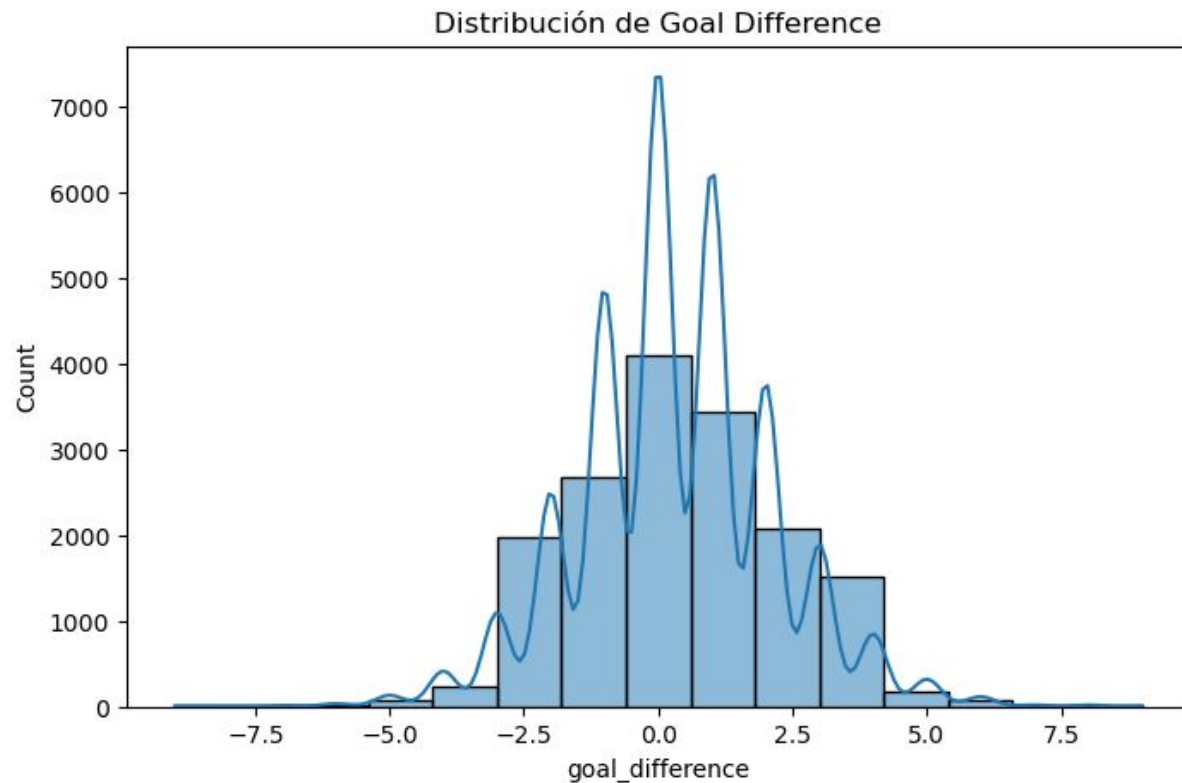
GRAPHICS Q1



GRAPHICS Q1



GRAPHICS Q1



Conclusion: Is xG a Good Predictor of Actual Goals?

Yes, with moderate-to-strong reliability.

- xG correlates **0.57** with total goals scored — a solid but imperfect predictor of scoring output.
- xG correlates **0.68** with goal difference — a stronger signal for predicting **match dominance** (who outplayed whom).
- **Key insight:** xG is better at showing *who created the better chances* than at predicting the *exact scoreline*. Finishing quality, goalkeeping, and in-game luck still account for meaningful variance.

Q2 - Does recent team performance influence match outcomes?

Relevant variables:

(from kaggle.com description of dataset)

- 13-14 - home team and away team current number of points just before the start of the match
- (not 15-16 - total number of points home/away team have gained in their previous 4 matches)
- 19-20 - total number of goals home/away team have scored in their previous 4 matches
- 23-24 - total number of goals home/away team have conceded (goals against) in their previous 4 matches
- (27-30 - average value of xG home/away team have gained in their previous 4 matches)

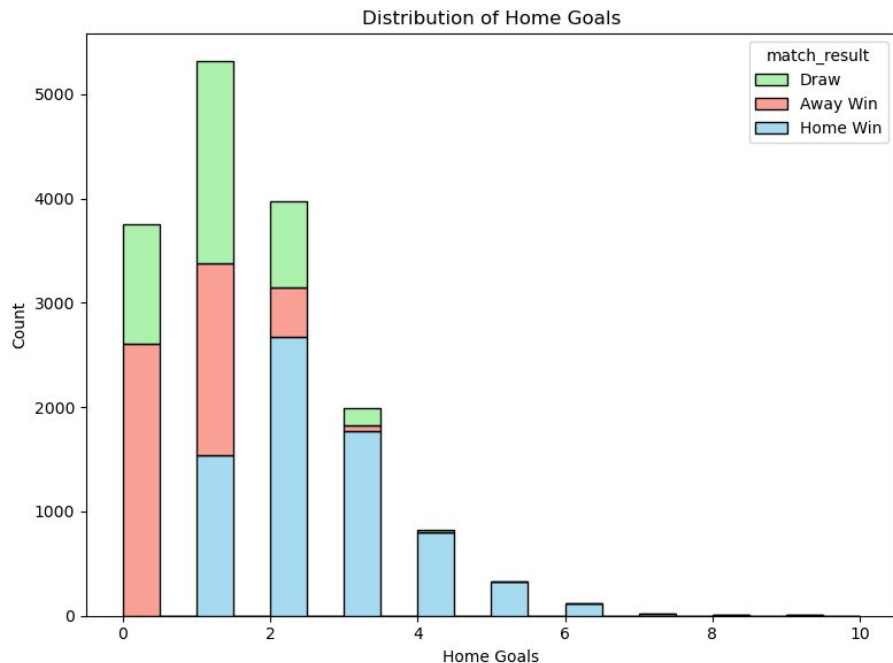
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---→

→ can match results be predicted?

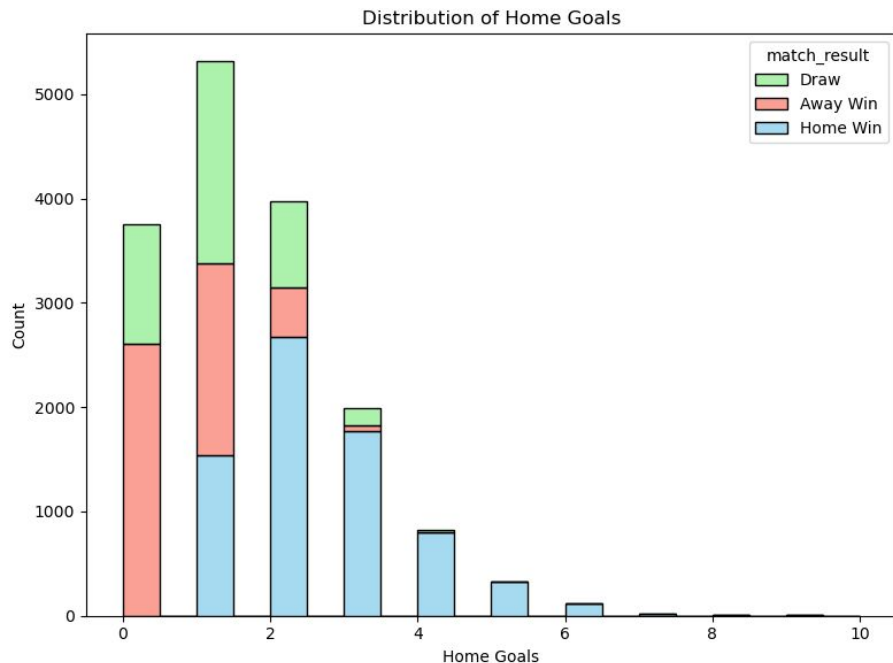
```
df["recent_form_difference"] = (df["home_points_last_4_matches"] - df["away_points_last_4_matches"])
```

GRAPHICS Q2: predict result with recent perf.



goals are almost categorical -
however: asking for
`match_result`
(in tendency: just win, draw, defeat)

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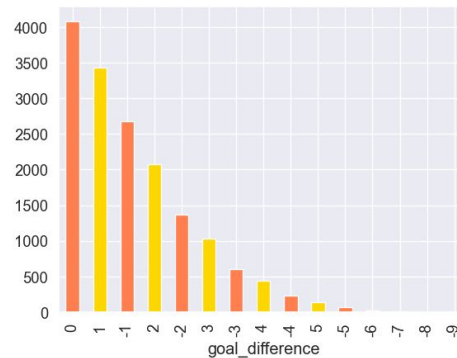


Table Q2: predict result with recent perf.

```
clean_df["match_result"].value_counts()['Home Win'] / total_matches,  
clean_df["match_result"].value_counts()['Draw'] / total_matches, clean_df["match_result"].value_counts()['Away  
Win'] / total_matches
```

The Probability of winning is 0.45 for Home Team.

The Probability of winning is 0.25 for Away Team.

The Probability of a Draw is 0.3.

Conclusion Q2: better than guess

crosstab (recent_perf_distant, match_result)

The recent form predicts

a Home Win with certainty

$$3719 / (1492 + 1652 + 3719) = 0.54$$

an Away Win with certainty

$$2872 / (2872 + 1895 + 2653) = 0.38$$

and a Draw with certainty

$$410 / (471 + 410 + 739) = 0.25$$

match_result	Away Win	Draw	Home Win
recent_form_difference_ten			
	140	130	159
Away Win	2872	1895	2653
Draw	471	410	739
Home Win	1492	1652	3719

Remember: The Probability of winning for arbitrary Home Team is 0.45. So “current_form” increased the likelihood of estimated result in this case! (decreased for Draws -0.05; increased - 0.25 → 0.38 - for away win)

xG: home win: 0.55; away win: 0.39; draw: 0.17

Q3: Do teams with more current points consistently achieve better match results?

Objective

Investigate whether teams with more accumulated league points **before a match** are more likely to achieve better results.

Hypothesis

Teams with higher league points before a match are expected to win more often.

Variables

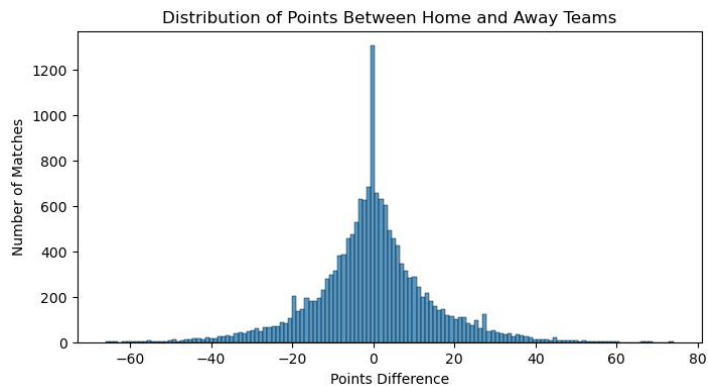
Primary variables

- Points Difference
- Match Result

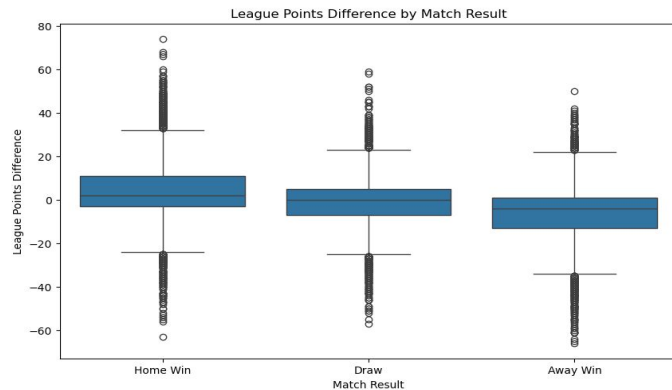
Supporting variables

- Goal Difference
- xG Difference

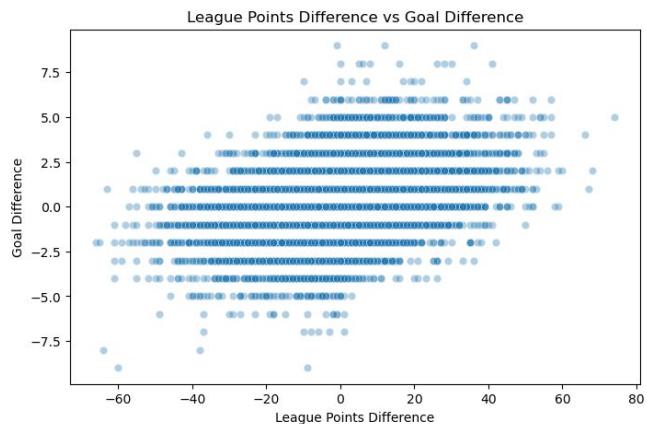
Histogram → Distribution



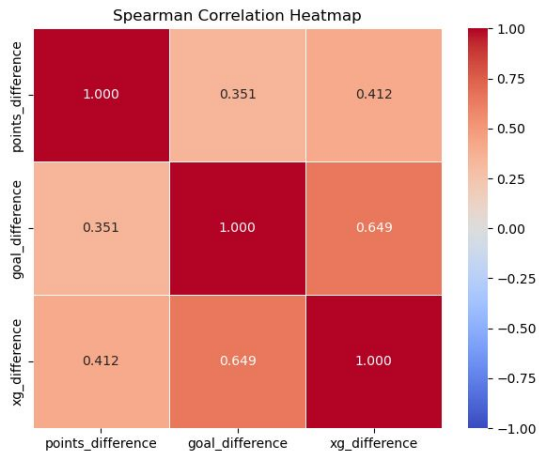
Boxplot → Comparing results



Scatterplot → Variables relationship



Heatmap → Spearman Correlations



Answer to: Do teams with more current points consistently achieve better match results?

League points matter.

Teams with higher league points generally achieve better match results .

Goal Difference Positive association $\rho = 0.351$

Higher point differences are associated with larger goal differences.

Expected Goals Positive association $\rho = 0.412$

League points show an even stronger relationship with expected goals.

Final Conclusion

Yes. Teams with higher accumulated league points generally achieve better match results .

Although they do not perfectly predict results, teams with higher accumulated points are consistently associated with better performances.

However, football remains unpredictable, so some lower-ranked teams may still obtain positive results.

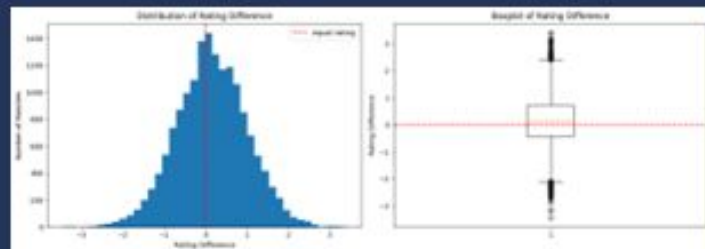
How do Team Player Ratings and Team Formations Influence Match Performance?

Key Variables:

- $\text{rating_difference} = \text{Home team avg rating} - \text{Away team avg rating}$
- $\text{home_formation} / \text{away_formation} = \text{Tactical setup (encoded 0-4)}$

1. Univariate Analysis — What Does the Data Look Like?

Rating Difference Distribution

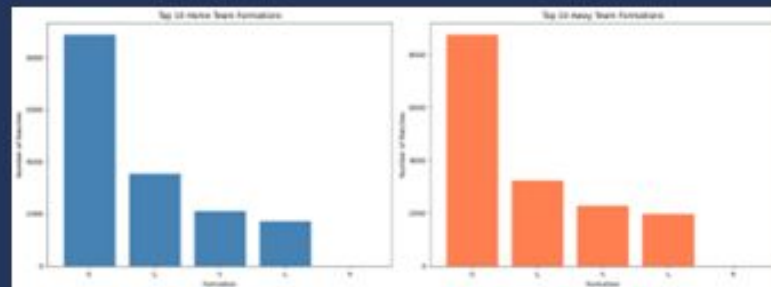


Key Numbers:

- Mean rating difference ≈ 0.00 — evenly matched
- Range: -5 to +5
- Symmetric, bell-shaped distribution

Formation Key: 0 = 2 def, 1 fwd (4-3-3) | 1 = 3 def, 1 fwd (3-4-2-1) | 2 = 2 def, 2 fwd (4-4-2) | 3 = 3 def, 2 fwd (3-5-2) | 4 = Other

Formation Distribution

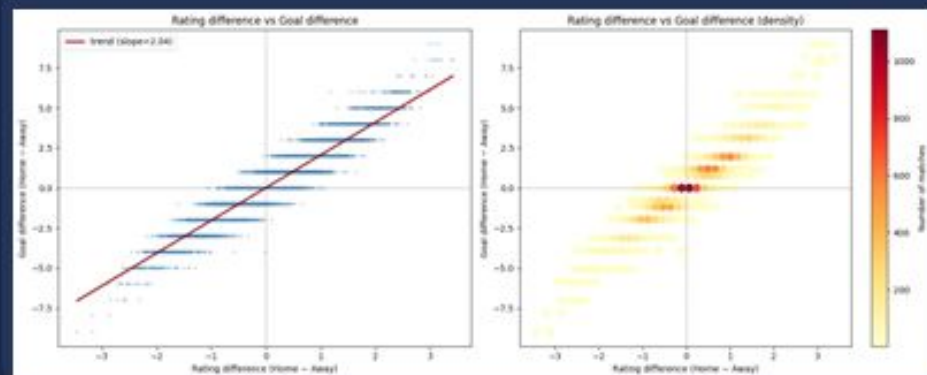


Key Numbers:

- Formation 0 (4-2-3-1 / 4-3-3) = $\sim 40\%$ of all matches
- Formation 2 (4-4-2) = $\sim 25\%$
- Home & away distributions nearly identical

2. Do Higher Ratings = More Goals?

Rating Difference vs Goal Difference



KEY STATISTICS

Correlation:
 $r \approx 0.50$

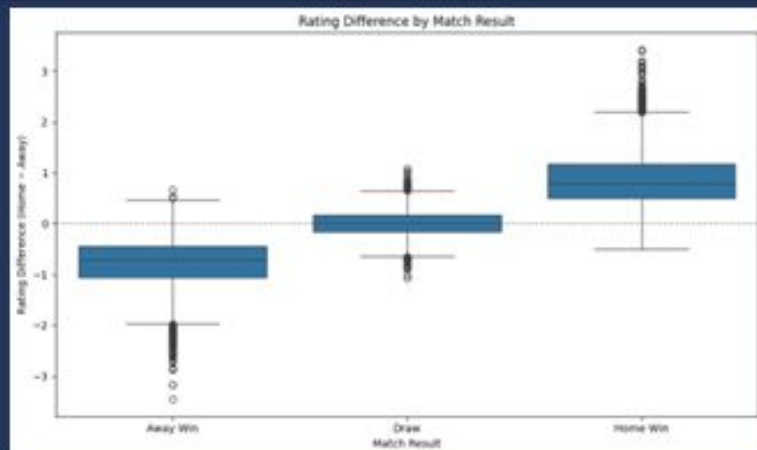
Trend line slope:
+0.44 goals per
+1 rating point

Highest density:
near (0, 0)
→ close matches
dominate

💡 In football analytics, $r = 0.50$ is actually quite strong — football is a low-scoring, high-variance sport.

3. Player Ratings Predict Match Outcomes

Rating Difference by Match Result



Match Result	Avg Rating Diff
--------------	-----------------

🏠 Home Win	+0.75
------------	-------

🤝 Draw	-0.05
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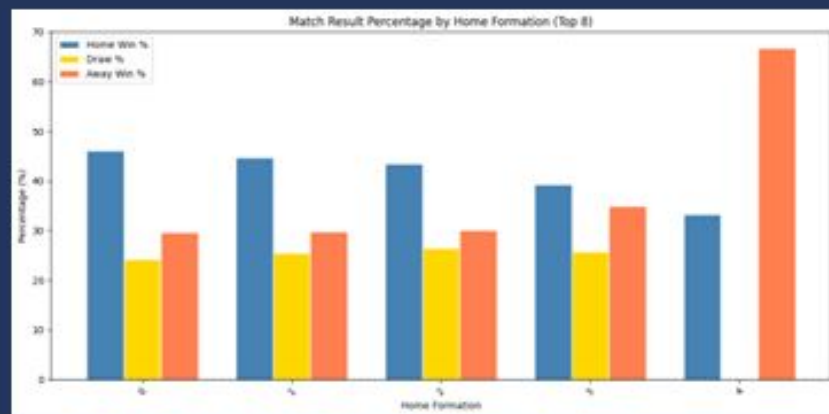
✈️ Away Win	-0.69
-------------	-------

💡 A perfect gradient:

Home rated higher → Home wins | Ratings equal → Draw | Away rated higher → Away wins

4. Do Formations Matter? — Barely

Match Result % by Home Formation (Top 8)



Key Numbers:

- Home win % across top 8 formations: 44–47%
- Variation between formations: ≤ 3 pct points
- Draw %: 25–28% across all formations
- Away win %: 27–29% across all formations

→ No formation is a magic bullet!

💡 Manchester City playing 4-4-2 beats a relegation side playing 4-3-3 every time.
Quality of the players matters far more than how you arrange them.

Final Answer

"Team player ratings are a strong predictor of match performance.
Team formations, by contrast, show only a weak association with results."

Three Key Takeaways

⚽ Ratings → Goals

$r = 0.50$ correlation
with goal difference

Each +1 rating $\approx$
+0.44 goals

⚽ Ratings → Wins

Clear gradient:
Away Win (-0.69)
→ Draw (0.00)
→ Home Win (+0.75)

⚽ Formations
Minor Impact

All formations produce
~45% home win rate

Quality > Shape

Combined with xG & recent form: Ratings = structural quality measure | xG & form = situational measures